

Trained

Place labels decide what counts



Large labelled training set

train



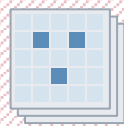
Corpus-specialized features

Standard
Strong within-domain discrimination

Cross-env.
Can inherit training-domain bias

Training-free, prior work

Added criteria decide what counts



Frozen patch features

Standard
Selection $\neq$ place discrimination

→



Codebook

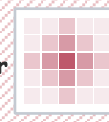
Prior-selected / aggregated features

Cross-env.
Selection criteria may not transfer



Segmentation CLS attention

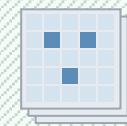
or



or

Ours

Let patches decide what counts



Frozen patch features

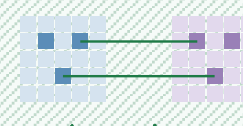
Standard
Match distinctive local evidence

→



Within-image

Balanced, matchable patch evidence

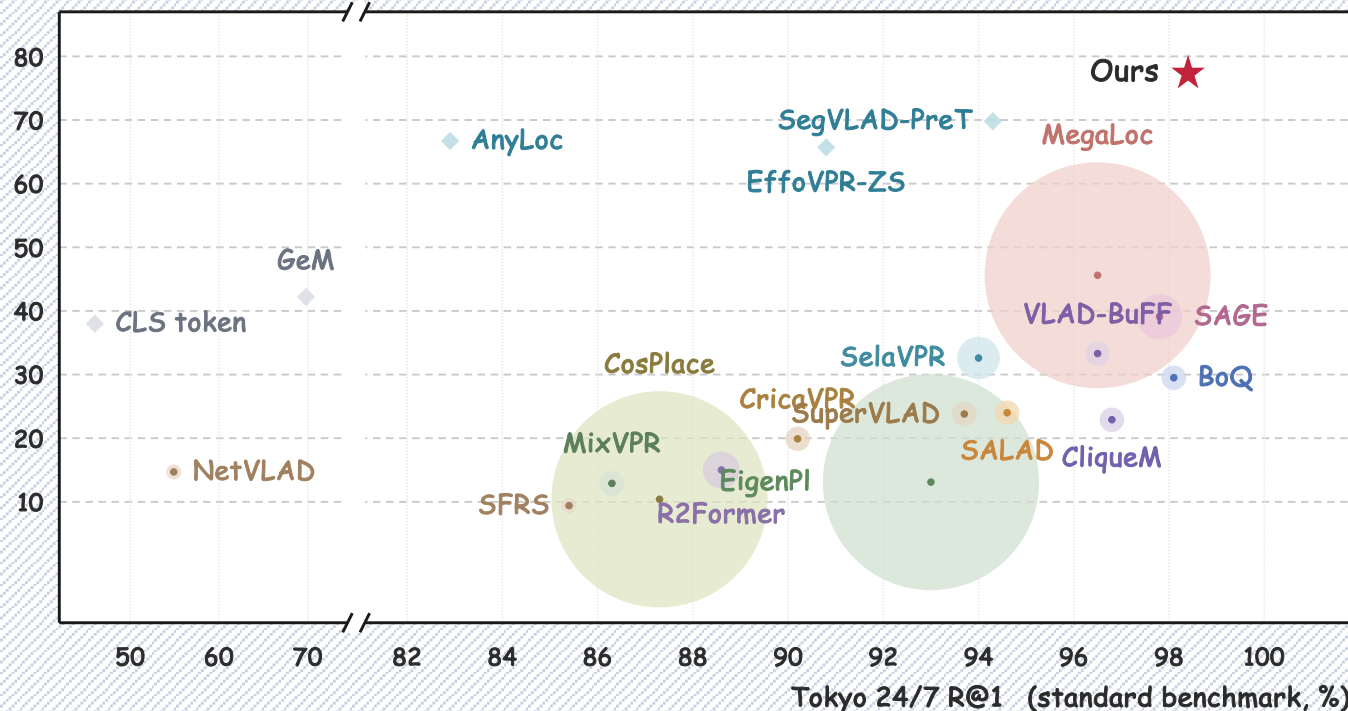


Across-image

Cross-env.
Strong cross-environment retrieval

Supervision cost vs. cross-environment recall

VP-Air R@1 (cross-environment benchmark, %)



★ ours (training-free) ♦ training-free ● trained ● 1 million images ● 5 million images

Recall across 17 benchmarks

